

MRI and non-MRI quantifiable neuroanatomical and functional parameters are useful for tractography

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Introduction

In the last two decades, diffusion magnetic resonance imaging (dMRI) - derived tractography has progressively become a cornerstone in mapping the brain's white matter pathways (Dell'Acqua, Descoteaux, & Leemans, 2024; Jeurissen, Descoteaux, Mori, & Leemans, 2017; Rheault, Poulin, Caron, St-Onge, & Descoteaux, 2020; Yeh, Jones, Liang, Descoteaux, & Connelly, 2020; Zhang et al., 2022). Nevertheless, white matter fiber tractography is an ill-posed process per definition, as there exists a multitude of possible fiber configurations that would represent the measured dMRI data equally well. Even if it were possible to map trajectories of white matter fibers' pathways with 100% accuracy, this would only tell us whether a road exists and not when or how it is "used" (Jones, 2010; Maffei et al., 2022; Maier-Hein et al., 2017; Schilling et al., 2019).

The integration of additional neuroanatomical parameters holds promise for addressing some of the limitations inherent to tractography (Dell'Acqua et al., 2024). For example, the axon diameter and dispersion of a bundle could inform on the complexity of its fibers' configuration (Andersson et al., 2020; Lee et al., 2019; Skoven, Andersson, Pizzolato, Siebner, & Dyrby, 2024) and as will be discussed in the Pro's section, could be of use to constrain the reconstruction algorithms in a useful manner. Anatomical priors or information derived from functional modalities such as functional MRI, electroencephalography, magnetoencephalography, or transcranial magnetic stimulation have previously been shown useful to obtain more complete and less spurious fiber reconstructions (Fotiadis et al., 2024; Rheault et al., 2019). Likewise, knowing the myelin content or the g-ratio, might help in either informing the tractography algorithms or validating the resulting quantitative connectivity estimation. Finally, these parameters could enhance the quantitative analysis of fiber tracts, providing hints on the physiological significance of fibers and potentially bridging the gap with biology (Drakesmith et al., 2019). However, it is not straightforward to properly estimate these parameters through MRI, while, at least, histology, light-sheet imaging, optical coherence tomography, and electron microscopy approaches represent a viable way to quantify them. Even though each of these neuroanatomical methods have intrinsic weaknesses; the main advantage is that they have much higher spatial resolution and, although they provide a static description of the tissues, by combining them, it could be possible to achieve a ground truth that can better inform the MRI tractography.

Debate

Here, we summarize the debate "Non-MRI quantifiable neuroanatomical parameters (such as axon density, axon dispersion, myelin) are useful for tractography." In the title, we have broadened this scope to also include MRI-derived parameters beyond those available through diffusion MRI (dMRI), such as T1, T2, functional MRI, and non-MRI metrics. This debate

examines the extent to which these additional quantifiable parameters can enhance MRI-based tractography by improving the accuracy of neural architecture reconstruction (Figure 1). We present the key arguments against this proposition (Con's) as well as those in support (Pro's), exploring the implications of incorporating multimodal data into tractographic frameworks.

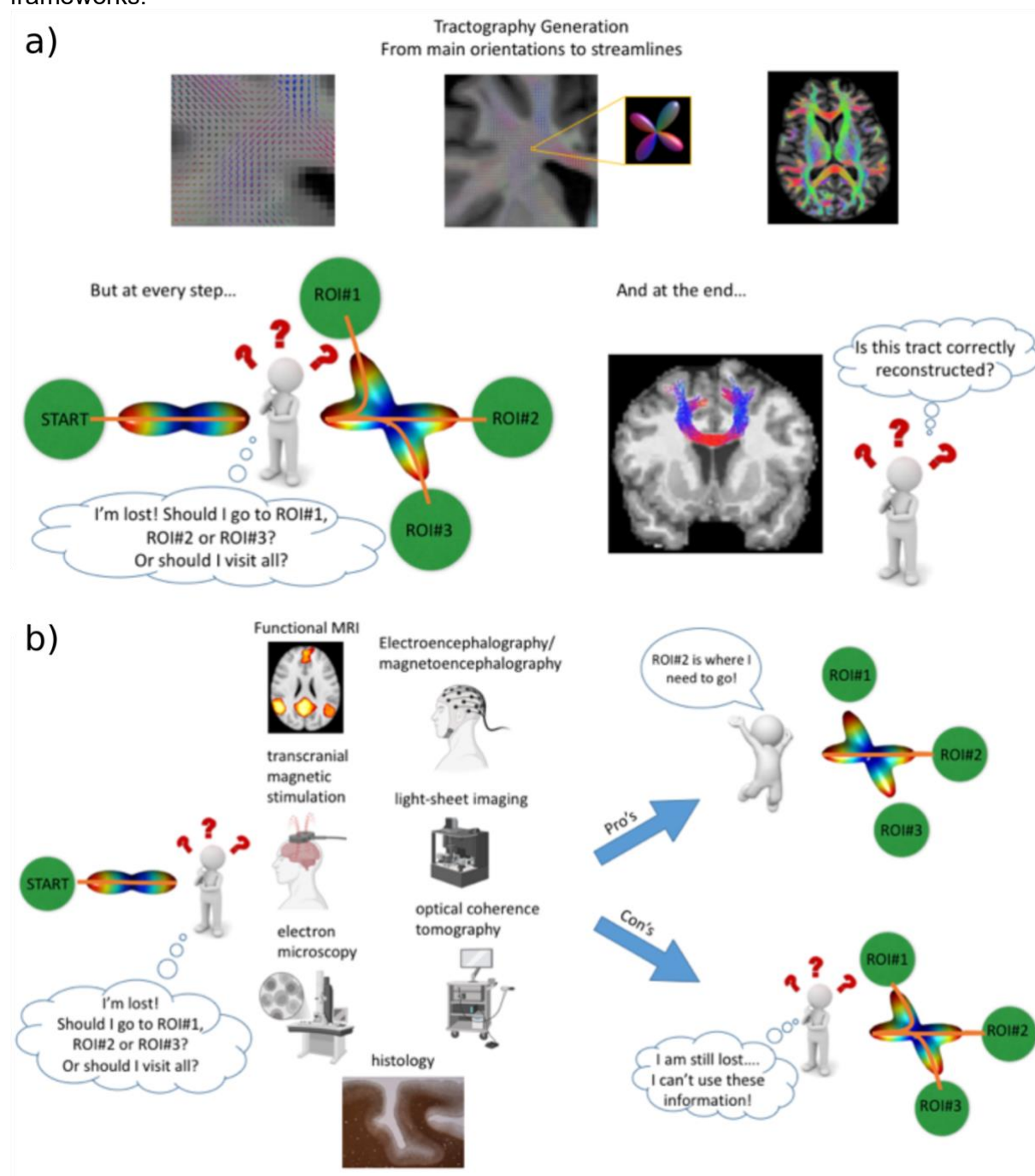


Figure 1: Graphical representation of the debate. On panel a), on top we show a standard tractography pipeline where from the principal diffusion directions we create the streamlines. On the bottom, we show two major issues of this processing: on the left, the fact that streamlines propagation encounters bottlenecks voxels where it is not clear which is the right path to follow; and, on the right, the problem of validation, that is once we have reconstructed a particular tract in-vivo there is no straightforward validation that assures that the specific bundle exists and is well reconstructed. On panel b), we resume the debate where we discuss if MRI and non-MRI quantifiable neuroanatomical and functional parameters are useful for tractography and, thus can be employed to solve the two issues stated above, or not. Some drawings of the figure were created in <https://BioRender.com>.

Con's

The evolution of tractography methodologies has progressively allowed reconstructed fibers' configurations to be combined with microstructural measures to understand how these parameters change along the tracts and provide a better representation of the tissues. However, many studies have identified significant challenges regarding the accurate representation of biological microstructural parameters such as axon diameters and myelin content using MRI signals. Current evidence suggests that parameters like myelin content or axonal diameters are not directly reflected in the MRI metrics (T1, T2, myelin water fraction, diffusion coefficient, etc.), which complicates the interpretation of results when comparing with anatomical knowledge (Afzali et al., 2021; Kiselev, 2021; Lee, Jespersen, Fieremans, & Novikov, 2020; Paquette, Eichner, Knösche, & Anwander, 2021). Thus, incorporating non-MRI-quantifiable parameters into tractography algorithms that only use MRI signals for which there is no direct mapping to such parameters, is likely to introduce additional uncertainty, rather than enhancing the accuracy of the results.

Moreover, histological measurements, while informative, are limited to post-mortem brain samples, creating barriers to translation to in vivo contexts. This limitation raises the question of relevance, as conducting tractography on postmortem brains has limited value for understanding living systems. Indeed, ex vivo analyses, though potentially rich in detail, cannot be easily extrapolated to in vivo scenarios, particularly in light of inter-individual variability and aging effects on neural architecture (how do we learn the transfer function from the age of the ex vivo sample to the in vivo patient?). Let us take as an example measurements of myelination derived from histology. Such measurements lack the granularity to assign unique myelination levels to individual axons and typically reflect a cross-sectional view rather than a comprehensive profile across the entirety of a fiber bundle which is what is ideally needed to be injected in tractography algorithms. Furthermore, the disparity in scale between MRI and other modalities complicates the integration of these data, making proper analysis non-trivial. Even when post-mortem tractography can be directly used for making correlation with histology, like in macaque studies (Girard et al., 2020; Yendiki et al., 2022), the number of samples employed is generally very low, which inevitably compromises the statistical relevance. In addition, the intermittent nature of myelination in healthy axons – not only in the nodes of Ranvier, but also in other parts of the axons (Auer, Vagionitis, & Czopka, 2018; Suminaite, Lyons, & Livesey, 2019). - further complicates the utility of these priors, as current tractography algorithms do not yet integrate this valuable information effectively and rather treat axons as families that follow common trajectories and, by necessity, average all these properties together.

Another important point to be considered is that acquiring additional MRI measurements beyond those strictly needed to perform tractography costs time (e.g., longer scans, or long post-processing analyses), while collecting non-MRI measurements not only costs time and money (e.g., long lab analyses, different equipment and specialized knowledge) but it might also end up being not specific to the individual - when invasive measures of the individual are not available. As such, they may add variability rather than help (e.g., population-based knowledge might not per se hold in the individual subject).

Finally, even by combining histology, staining, tracers etc., or complex functional measures like “conduction velocities” or TMS data with tractography, obtaining precise measurements of a variety of human subjects remains particularly challenging, as extensive, and complex protocols must be implemented, once again introducing additional uncertainty rather than clarity. Furthermore, the reliance on non-MRI methods can exacerbate these complexities, as their accuracy alone cannot compensate for the inherent difficulties within tractography methodologies. As the complexity of our knowledge increases, so too do the corresponding challenges, leading to a paradox where deeper understanding may generate hesitancy rather than confidence in the application of advanced tractography techniques.

Pro's

The community has already shown that the inclusion of non-diffusion MRI data, such as functional magnetic resonance imaging, can enhance anatomical accuracy in tractography (Dell'Acqua et al., 2024). More in detail, knowing how fast the functional information is transmitted along with geometrical properties of fibers like the g-ratio can enlighten on which reconstructed trajectories are used by the brain and thus consist in true positive reconstruction. The integration of post-mortem representations of neural tracts can also serve as valuable feedback for improving tractography methodologies. For example, by utilizing discriminative models that incorporate anatomical information as priors - such as bundles' myelination - researchers can refine the optimization process by either penalizing or promoting certain bundles more than others in the connectivity estimations as shown for example in (Daducci, Schiavi, Christiaens, Smith, & Alexander, 2024; Schiavi et al., 2020). Furthermore, insights drawn from cortical histology may provide critical information about bundles terminations and their microstructural properties that could enhance tractography, potentially informing the accuracy of trajectories in neural modeling. Indeed, recent advancements using electron microscopy technique have shown that it is now possible to segment whole axons and thus directly measure microstructural properties such as myelin and diameter along their path opening new perspectives of finding a way to follow entire bundles of axons (Karlupia et al., 2023; Kjer et al., 2024; Leckenby, Chacon, Grobbelaar, & Lichtman, 2019).

Histological investigation also offers a deeper understanding of axonal properties, revealing not only microstructural properties like myelination, axon diameters and dispersions, but also interesting phenomena such as axons bending at sharp angles (even 90 degrees!) or splitting into two, which were previously unknown to the tractography community, with the risk of introducing inaccurate strong priors on the algorithms (Catani, Bodi, & Dell'Acqua, 2012; Wedeen et al., 2012a, 2012b). Combining geometrical priors and the conservation of microstructural properties along brain tracts as new constraints for the algorithms is a potential key factor to enhance the accuracy of tractography techniques. Moreover, at the moment, post-mortem histology surely offers a unique way to validate the developed methods and learn how to improve their quality on ground truth data that are more representative of tissue complexity than synthetic phantoms.

Finally, validation against biopsy and/or cerebral samples from neurosurgical patients to understand where tractography succeeds and fails in modeling specific diseases can guide researchers in refining their approaches, paving the way for more effective clinical applications. Indeed, the validation of diffusion and tract metrics through for example in vivo direct electrical stimulations can facilitate their integration into clinical practices such as neurosurgery, ensuring that the tractography outputs are robust and clinically relevant (Radwan et al., 2024).

Closing Statements

Despite challenges posed in the con's paragraph, every crisis is an opportunity and getting diverse types of histological and functional information is an opportunity to learn the transfer function between different imaging modalities and biological realities. This can enhance our understanding of complex neural architectures and improve the accuracy of tractography models. Ultimately, overcoming these challenges can lead to significant benefits in both research and clinical practices by refining methodologies and contributing to more effective applications in neuroscience and medicine.

References

Afzali, M., Pieciak, T., Newman, S., Garyfallidis, E., Özarslan, E., Cheng, H., & Jones, D. K. (2021). The sensitivity of diffusion MRI to microstructural properties and experimental factors. *Journal of Neuroscience Methods*.
<http://doi.org/10.1016/j.jneumeth.2020.108951>

- Andersson, M., Kjer, H. M., Rafael-Patino, J., Pacureanu, A., Pakkenberg, B., Thiran, J. P., ... Dyrby, T. B. (2020). Axon morphology is modulated by the local environment and impacts the noninvasive investigation of its structure–function relationship. *Proceedings of the National Academy of Sciences of the United States of America*, 117(52), 33649–33659. <http://doi.org/10.1073/PNAS.2012533117>
- Auer, F., Vagionitis, S., & Czopka, T. (2018). Evidence for Myelin Sheath Remodeling in the CNS Revealed by In-Vivo Imaging. *Current Biology*, 28(4), 549–559.e3. JOUR. <http://doi.org/10.1016/j.cub.2018.01.017>
- Catani, M., Bodi, I., & Dell'Acqua, F. (2012). Comment on “The Geometric Structure of the Brain Fiber Pathways.” *Science*, 337(6102), 1605. article. <http://doi.org/10.1126/science.1223425>
- Daducci, A., Schiavi, S., Christiaens, D., Smith, R., & Alexander, D. C. (2024). Global tractography. In A. Dell'Acqua, Flavio and Descoteaux, Maxime and Leemans (Ed.), *Handbook of Diffusion MR Tractography* (1st ed.). Elsevier. <http://doi.org/https://doi.org/10.1016/B978-0-12-818894-1.00014-8>
- Dell'Acqua, F., Descoteaux, M., & Leemans, A. (2024). *Handbook of Diffusion MR Tractography: Imaging Methods, Biophysical Models, Algorithms and Applications* (1st ed.). book, Academic Press. Retrieved from <https://books.google.it/books?id=IJruEAAAQBAJ>
- Drakesmith, M., Harms, R., Rudrapatna, S. U., Parker, G. D., Evans, C. J., & Jones, D. K. (2019). Estimating axon conduction velocity in vivo from microstructural MRI. *NeuroImage*, 203, 116186. article. <http://doi.org/https://doi.org/10.1016/j.neuroimage.2019.116186>
- Fotiadis, P., Parkes, L., Davis, K. A., Satterthwaite, T. D., Shinohara, R. T., & Bassett, D. S. (2024). Structure–function coupling in macroscale human brain networks. *Nature Reviews Neuroscience*, 25(10), 688–704. JOUR. <http://doi.org/10.1038/s41583-024-00846-6>
- Girard, G., Caminiti, R., Battaglia-Mayer, A., St-Onge, E., Ambrosen, K. S., Eskildsen, S. F., ... Innocenti, G. M. (2020). On the cortical connectivity in the macaque brain: A comparison of diffusion tractography and histological tracing data. *NeuroImage*, 221, 117201. article. <http://doi.org/https://doi.org/10.1016/j.neuroimage.2020.117201>
- Jeurissen, B., Descoteaux, M., Mori, S., & Leemans, A. (2017). Diffusion MRI fiber tractography of the brain. *NMR in Biomedicine*, e3785. <http://doi.org/10.1002/nbm.3785>
- Jones, D. K. (2010). Challenges and limitations of quantifying brain connectivity in vivo with diffusion MRI. *Imaging in Medicine*. <http://doi.org/10.2217/iim.10.21>
- Karlupia, N., Schalek, R. L., Wu, Y., Meirovitch, Y., Wei, D., Charney, A. W., ... Lichtman, J. W. (2023). Immersion Fixation and Staining of Multicubic Millimeter Volumes for Electron Microscopy–Based Connectomics of Human Brain Biopsies. *Biological Psychiatry*, 94(4), 352–360. article. <http://doi.org/https://doi.org/10.1016/j.biopsych.2023.01.025>
- Kiselev, V. G. (2021). Microstructure with diffusion MRI: what scale we are sensitive to? *Journal of Neuroscience Methods*. <http://doi.org/10.1016/j.jneumeth.2020.108910>
- Kjer, H. M., Andersson, M., He, Y., Pacureanu, A., Daducci, A., Pizzolato, M., ... Dyrby, T. B. (2024). Bridging the 3D geometrical organisation of white matter pathways across anatomical length scales and species. article. <http://doi.org/10.7554/elife.94917.2>
- Leckenby, J. I., Chacon, M. A., Grobbelaar, A. O., & Lichtman, J. W. (2019). Imaging Peripheral Nerve Regeneration: A New Technique for 3D Visualization of Axonal Behavior. *Journal of Surgical Research*, 242, 207–213. article. <http://doi.org/https://doi.org/10.1016/j.jss.2019.04.046>
- Lee, H. H., Jespersen, S. N., Fieremans, E., & Novikov, D. S. (2020). The impact of realistic axonal shape on axon diameter estimation using diffusion MRI. *NeuroImage*, 223. <http://doi.org/10.1016/j.neuroimage.2020.117228>
- Lee, H. H., Yaros, K., Veraart, J., Pathan, J. L., Liang, F. X., Kim, S. G., ... Fieremans, E. (2019). Along-axon diameter variation and axonal orientation dispersion revealed with 3D electron microscopy: implications for quantifying brain white matter microstructure

- with histology and diffusion MRI. *Brain Structure and Function*.
<http://doi.org/10.1007/s00429-019-01844-6>
- Maffei, C., Girard, G., Schilling, K. G., Aydogan, D. B., Adluru, N., Zhylka, A., ... Yendiki, A. (2022). Insights from the IronTract challenge: Optimal methods for mapping brain pathways from multi-shell diffusion MRI. *NeuroImage*, 257.
<http://doi.org/10.1016/j.neuroimage.2022.119327>
- Maier-Hein, K. H., Neher, P. F., Houde, J.-C., Côté, M.-A., Garyfallidis, E., Zhong, J., ... Descoteaux, M. (2017). The challenge of mapping the human connectome based on diffusion tractography. *Nature Communications*, 8(1), 1349. article.
- Paquette, M., Eichner, C., Knösche, T. R., & Anwender, A. (2021). Axon Diameter Measurements using Diffusion MRI are Infeasible. *bioRxiv*, 2020.10.01.320507.
<http://doi.org/https://doi.org/10.1101/2020.10.01.320507>
- Radwan, A. M., Emsell, L., Vansteelandt, K., Cleeren, E., Peeters, R., De Vleeschouwer, S., ... Sunaert, S. (2024). Comparative validation of automated presurgical tractography based on constrained spherical deconvolution and diffusion tensor imaging with direct electrical stimulation. *Human Brain Mapping*, 45(6), e26662. article.
<http://doi.org/https://doi.org/10.1002/hbm.26662>
- Rheault, F., Poulin, P., Caron, A. V., St-Onge, E., & Descoteaux, M. (2020). Common misconceptions, hidden biases and modern challenges of {dMRI} tractography. *Journal of Neural Engineering*, 17(1), 11001. article. <http://doi.org/10.1088/1741-2552/ab6aad>
- Rheault, F., St-Onge, E., Sidhu, J., Maier-Hein, K., Tzourio-Mazoyer, N., Petit, L., & Descoteaux, M. (2019). Bundle-specific tractography with incorporated anatomical and orientational priors. *NeuroImage*, 186, 382–398. article.
<http://doi.org/https://doi.org/10.1016/j.neuroimage.2018.11.018>
- Schiavi, S., Ocampo-Pineda, M., Barakovic, M., Petit, L., Descoteaux, M., Thiran, J. P., & Daducci, A. (2020). A new method for accurate in vivo mapping of human brain connections using microstructural and anatomical information. *Science Advances*.
<http://doi.org/10.1126/sciadv.aba8245>
- Schilling, K. G. K. G., Nath, V., Hansen, C., Parvathaneni, P., Blaber, J., Gao, Y., ... Landman, B. A. B. A. (2019). Limits to anatomical accuracy of diffusion tractography using modern approaches. *NeuroImage*, 185, 1–11.
<http://doi.org/10.1016/j.neuroimage.2018.10.029>
- Skoven, C. S., Andersson, M., Pizzolato, M., Siebner, H. R., & Dyrby, T. B. (2024). Mapping axon diameters and conduction velocity in the rat brain {textendash} different methods tell different stories of the structure-function relationship. *bioRxiv*. article.
<http://doi.org/10.1101/2023.10.20.558833>
- Suminaite, D., Lyons, D. A., & Livesey, M. R. (2019). Myelinated axon physiology and regulation of neural circuit function. *Glia*, 67(11), 2050–2062. article.
<http://doi.org/https://doi.org/10.1002/glia.23665>
- Wedeen, V. J., Rosene, D. L., Wang, R., Dai, G., Mortazavi, F., Hagmann, P., ... Tseng, W.-Y. I. (2012a). Response to Comment on “The Geometric Structure of the Brain Fiber Pathways.” *Science*, 337(6102), 1605. article. <http://doi.org/10.1126/science.1223493>
- Wedeen, V. J., Rosene, D. L., Wang, R., Dai, G., Mortazavi, F., Hagmann, P., ... Tseng, W.-Y. I. (2012b). The Geometric Structure of the Brain Fiber Pathways. *Science*, 335(6076), 1628–1634. article. <http://doi.org/10.1126/science.1215280>
- Yeh, C.-H., Jones, D. K., Liang, X., Descoteaux, M., & Connelly, A. (2020). Mapping Structural Connectivity Using Diffusion MRI: Challenges and Opportunities. *Journal of Magnetic Resonance Imaging*, n/a(n/a). article. <http://doi.org/10.1002/jmri.27188>
- Yendiki, A., Aggarwal, M., Axer, M., Howard, A. F. D., van Cappellen van Walsum, A.-M., & Haber, S. N. (2022). Post mortem mapping of connectional anatomy for the validation of diffusion MRI. *NeuroImage*, 256, 119146. article.
<http://doi.org/https://doi.org/10.1016/j.neuroimage.2022.119146>
- Zhang, F., Daducci, A., He, Y., Schiavi, S., Seguin, C., Smith, R. E., ... O'Donnell, L. J. (2022). Quantitative mapping of the brain's structural connectivity using diffusion MRI

tractography: A review. *NeuroImage*, 249, 118870. misc.
<http://doi.org/https://doi.org/10.1016/j.neuroimage.2021.118870>